

# Paper 1

## Paper Information

- Title: What Does This Acronym Mean? Introducing a New Dataset for Acronym Identification and Disambiguation
- Authors: Amir Pouran Ben Veyseh, Franck Dernoncourt, Quan Hung Tran, Thien Huu Nguyen
- Year: 2020
- Journal/Conference: Proceedings of the 28th International Conference on Computational Linguistics (COLING 2020); also arXiv:2010.14678
- DOI: Not Mentioned

## Summary

- Research Problem: Existing datasets for Acronym Identification (AI) and Acronym Disambiguation (AD) are either too small, noisy (created by rule-based heuristics), or limited mostly to the medical domain, which hinders training of high-performing deep learning models for these tasks in the scientific domain.
- Objective: To build large, manually annotated datasets for AI and AD in the scientific domain, and to propose a strong deep learning baseline model for AD.
- Motivation: Acronyms are widely used in scientific writing (about 15% of PubMed queries and many clinical notes contain abbreviations); correctly identifying and disambiguating them is essential for downstream NLP tasks like question answering, slot-filling, and definition extraction.
- Method: Collected 6,786 English arXiv papers, selected candidate sentences likely to contain acronyms, and had Amazon Mechanical Turk workers manually annotate short forms and long forms (for AI). Used these annotations to build a dictionary of ambiguous acronyms and automatically generated an AD dataset using the "one-sense-per-discourse" assumption. Proposed a new model called GAD (Graph-based Acronym Disambiguation) that combines BiLSTM sequence encoding with a Graph Convolutional Network (GCN) over the sentence's dependency tree to capture long-range syntactic context.
- Dataset: SciAI (17,506 sentences, for AI) and SciAD (62,441 samples, for AD); also compared against UAD (Wikipedia-based) and a scientific variant SciUAD.
- Algorithms: LSTM-CRF (for AI); BiLSTM + GCN over dependency tree (GAD model) for AD; compared against NOA, ADE, UAD, BIOADI, LNCRF (AI baselines) and MF, ADE, NOA, UAD, BEM, DECBAE (AD baselines).
- Evaluation Metrics: Precision, Recall, Macro-averaged F1 score.
- Results: LSTM-CRF achieved 86.55 Macro F1 on SciAI (human performance 96.09). GAD achieved 81.90 Macro F1 on SciAD, outperforming all baselines (human performance 96.10).
- Key Findings: Scientific-domain acronym disambiguation (SciAD) is more challenging than general domain (UAD) due to longer sentences and higher semantic similarity between long forms. Syntactic dependency structure helps disambiguate acronyms that depend on distant words in a sentence.
- Advantages: Large, human-annotated, high-quality datasets; syntactic structure improves disambiguation for long-distance dependencies.
- Limitations: All models still perform far below human-level accuracy; annotation errors exist (e.g., popular acronyms, embedded acronyms, multiple acronyms in a sentence).
- Future Work: Developing more advanced models to close the gap with human performance.
- Applications: Text understanding, question answering, document retrieval, definition extraction.

## Keywords

- Author Keywords: Not Mentioned (not explicitly listed)

- Technical Keywords: Acronym Identification, Acronym Disambiguation, Sequence Labeling, Graph Convolutional Network, BiLSTM, Dependency Tree, SciAI, SciAD

### Technical Concepts

- Machine Learning / Deep Learning Models: BiLSTM for sequence encoding; Graph Convolutional Network (GCN) for structural/syntactic encoding; CRF layer for sequence labeling.
- Architecture or Framework: Sentence Encoder (BiLSTM) → Context Encoder (GCN over dependency tree) → Prediction (feed-forward classifier).
- Feature Extraction: Pre-trained word embeddings + POS tag embeddings; dependency parse trees.
- Data Preprocessing: Sentence filtering based on capital-letter heuristics to find acronym candidates; Levenshtein distance-based normalization of long forms.
- Training Method: Supervised training with negative log-likelihood loss (AI) and classification loss (AD); Adam optimizer.
- Evaluation Process: 80:10:10 train/dev/test split; macro-averaged P/R/F1.

### Student Notes

- What is the paper mainly about? Creating large annotated datasets (SciAI, SciAD) for identifying and disambiguating acronyms in scientific papers, plus a new GCN-based disambiguation model (GAD).
- What problem does it solve? Lack of large, high-quality, domain-diverse datasets for acronym tasks in the scientific literature.
- Why is this research important? Acronyms are everywhere in scientific text; understanding them correctly is key for automated text understanding systems.
- Main contributions: (1) SciAI and SciAD datasets, (2) GAD model using dependency trees, (3) extensive benchmarking against existing methods.
- What to remember: GAD's key idea—using syntactic dependency trees via GCN—helps capture context for acronyms whose meaning depends on distant words.

### Easy Explanation

Scientists use lots of short forms (like "CNN" for Convolutional Neural Network) in their papers. Computers need to find these short forms and figure out what they mean. The researchers created two new big datasets: one for finding acronyms in text (SciAI) and one for figuring out what an ambiguous acronym means in context (SciAD). They also built a new AI model called GAD that looks at how words are grammatically connected in a sentence (like a tree) to better guess an acronym's true meaning, especially when the clue word is far away in the sentence.

### One Paragraph Summary

This paper addresses limitations in existing acronym identification (AI) and acronym disambiguation (AD) datasets, which are often small, noisy, or restricted to the medical domain. The authors built two large, manually annotated datasets from 6,786 arXiv papers: SciAI (17,506 sentences for identifying acronyms and their long forms) and SciAD (62,441 samples for disambiguating ambiguous acronyms). They also proposed GAD, a new disambiguation model combining BiLSTM sequence encoding with Graph Convolutional Networks over dependency trees to capture long-range syntactic relationships between an acronym and its context. Experiments show that existing state-of-the-art models, including LSTM-CRF for AI and various baselines for AD, perform well below human-level accuracy on these harder, more realistic datasets. GAD outperformed all AD baselines, demonstrating the value of syntactic structure, though a significant gap remains versus human performance (96%+ F1). The datasets and model provide a stronger, more challenging benchmark for future acronym-understanding research and support downstream tasks such as question answering and document retrieval.

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# Paper 2

## Paper Information

- Title: AT-BERT: Adversarial Training BERT for Acronym Identification (Winning Solution for SDU@AAAI-21)
- Authors: Danqing Zhu, Wangli Lin, Yang Zhang, Qiwei Zhong, Guanxiong Zeng, Weilin Wu, Jiayu Tang
- Year: 2021
- Journal/Conference: AAAI-21 Workshop on Scientific Document Understanding (SDU); also arXiv:2101.03700 and CEUR-WS Vol-2831 proceedings
- DOI: Not Mentioned

## Summary

- Research Problem: Small size of manually annotated acronym identification datasets limits the training of robust deep learning models, causing overfitting.
- Objective: To build a robust, high-performing acronym identification system that won the SDU@AAAI-21 shared task.
- Motivation: Pre-trained language models like BERT can capture better semantic representations but need robustness improvements against overfitting on small datasets.
- Method: Fine-tune multiple pre-trained BERT-style models (BERT, SciBERT, RoBERTa, ALBERT, ELECTRA) for sequence labeling; incorporate FGM (Fast Gradient Method) adversarial training to perturb input embeddings during fine-tuning for robustness; ensemble multiple BERT variants by averaging output probabilities.
- Dataset: SciAI dataset (from Veyseh et al. 2020), used via SDU@AAAI-21 shared task splits (14,006 train / 1,717 dev / 1,750 test).
- Algorithms: BERT\_BASE, BERT\_LARGE, SciBERT, RoBERTa, ALBERT, ELECTRA; FGM adversarial training; model ensembling (BERT-E, AT-BERT-E).
- Evaluation Metrics: Precision, Recall, F1 (Macro F1) for acronym and long-form boundary prediction.
- Results: AT-BERT-E (ensemble with adversarial training) achieved 94.12 Macro F1, best among all compared methods; improved 8.66% and 7.57% over rule-based and LSTM-CRF baselines respectively. Human performance is 96.09.
- Key Findings: Adversarial training (FGM) consistently improves BERT-based models' robustness and F1 score; ensembling multiple BERT variants further boosts performance.
- Advantages: State-of-the-art results; robust to noisy/small data; won first place in the shared task.
- Limitations: Still below human-level performance; increased training time (roughly 2x) due to adversarial training.
- Future Work: Explore other adversarial training strategies (PGD, FreeLB); try alternative loss functions (Dice Loss, Focal Loss) to address class imbalance.
- Applications: Scientific document understanding, automated glossary generation, text mining.

## Keywords

- Author Keywords: Not Mentioned
- Technical Keywords: Acronym Identification, BERT, Adversarial Training, FGM, Model Ensemble, SciBERT, RoBERTa, ELECTRA, ALBERT

## Technical Concepts

- Machine Learning / Deep Learning Models: BERT and its variants (SciBERT, RoBERTa, ALBERT, ELECTRA) fine-tuned for token classification.
- Architecture or Framework: Pre-trained Transformer encoder + dense/softmax classification layer + Cross-Entropy loss; adversarial perturbation added to embeddings.

- Feature Extraction: Contextual token embeddings from BERT-family models.
- Data Preprocessing: BIO tagging scheme (B-short, I-short, B-long, I-long, O).
- Training Method: Supervised fine-tuning with FGM adversarial perturbation (gradient-based noise added to embeddings to increase robustness), Cross-Entropy loss.
- Evaluation Process: Macro F1 computed on test set for short-form and long-form boundary detection.

### Student Notes

- What is the paper mainly about? Using BERT with adversarial training and model ensembling to identify acronyms more accurately.
- What problem does it solve? Overfitting and limited robustness of BERT models trained on small acronym identification datasets.
- Why is this research important? It shows how simple techniques (adversarial training, ensembling) can significantly boost performance on a real shared-task benchmark.
- Main contributions: First use of adversarial training with BERT for acronym identification; multi-BERT ensemble winning the AAAI-21 shared task.
- What to remember: FGM adds small perturbations to word embeddings during training to make the model less sensitive to noise, improving generalization.

### Easy Explanation

The authors used a powerful language model (BERT) to find acronyms and their meanings in text, similar to highlighting important words. Since the training data was small, the model could "memorize" too much and fail on new text. To fix this, they added tiny random "noise" (perturbations) to the input during training - this is called adversarial training - so the model learns to be more robust. They also combined predictions from multiple different BERT-style models (like averaging opinions from several experts) to get more accurate results. This approach won first place in a major AI competition.

### One Paragraph Summary

This paper presents AT-BERT, the winning solution for the SDU@AAAI-21 acronym identification shared task. The authors address the challenge of limited training data by fine-tuning several BERT-family pre-trained models (BERT, SciBERT, RoBERTa, ALBERT, ELECTRA) as sequence labelers, then applying FGM-based adversarial training to make the models more robust against overfitting by injecting small perturbations into word embeddings during fine-tuning. Finally, they ensemble the four best-performing adversarially-trained models (AT-BERT-E) by averaging output probabilities. Experiments on the SciAI dataset show that all BERT-based models outperform rule-based and LSTM-CRF baselines, that adversarial training consistently boosts each model's F1 score, and that the final ensemble achieves a Macro F1 of 94.12%, substantially higher than the baseline (85.46%) though still short of human performance (96.09%). This work demonstrates the effectiveness of combining pre-trained transformers, adversarial training, and ensembling for robust acronym identification in scientific text.

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## Paper 3

### Paper Information

- Title: MadDog: A Web-based System for Acronym Identification and Disambiguation
- Authors: Amir Pouran Ben Veyseh, Franck Dernoncourt, Walter Chang, Thien Huu Nguyen
- Year: 2021
- Journal/Conference: Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2021), System Demonstrations; also arXiv:2101.09893

- DOI: Not Mentioned

## Summary

- Research Problem: No existing publicly available system can identify and disambiguate acronyms across multiple domains (scientific, biomedical, general, financial) in a unified way.
- Objective: To build the first web-based, multi-domain acronym identification and disambiguation system.
- Motivation: Acronyms are hard to understand, especially when not defined locally in the text or when ambiguous; existing tools are domain-limited or unavailable publicly.
- Method: Two-component system: (1) Acronym Identification using an extended, prioritized rule-based approach (Acronym Detector, Bounded Schwartz's, Character Match, Initial Capitals rules); (2) Acronym Expansion using a large Diverse acRonym Glossary (DOG) built from multiple domain corpora, plus a BiLSTM-based deep learning model trained on a massive automatically-labeled dataset (MAD, 46 million records) to disambiguate ambiguous acronyms based on context.
- Dataset: DOG glossary (426,389 acronyms, 3,781,739 long forms) built from Wikipedia, arXiv, Reddit, Medline abstracts, and PMC OA subset; MAD dataset (46 million records) for training the disambiguation model; evaluated on SciAI and SciAD benchmarks.
- Algorithms: Rule-based AI model (extended Schwartz and Hearst rules); BiLSTM + max-pooling + 2-layer feed-forward classifier for AD.
- Evaluation Metrics: Precision, Recall, F1, Macro F1.
- Results: On SciAI: MadDog achieved 88.12 Macro F1, outperforming rule-based and feature-based baselines. On SciAD: MadDog achieved 88.49 F1, best among compared models (GAD achieved 81.90).
- Key Findings: A rule-based AI approach with additional refined rules can outperform even some deep learning models; using a massive automatically-labeled disambiguation dataset (MAD) with 46M records improves generalization across many acronyms and domains.
- Advantages: Multi-domain support, publicly available web system, high recall while retaining rule-based precision.
- Limitations: Model requires large disk space (64 GB) and GPU/CPU resources; not evaluated as extensively for domains outside the benchmarks.
- Future Work: Integration into e-readers so users can quickly look up acronym meanings while reading.
- Applications: Reading assistance for technical documents, e-readers, information extraction, definition extraction, question answering.

## Keywords

- Author Keywords: Not Mentioned
- Technical Keywords: Acronym Identification, Acronym Disambiguation, Rule-based System, BiLSTM, Diverse acRonym Glossary (DOG), Massive Acronym Disambiguation (MAD) dataset, Web-based System

## Technical Concepts

- Machine Learning / Deep Learning Models: BiLSTM network for encoding sentence context; feed-forward neural classifier for long-form prediction.
- Architecture or Framework: Rule-based pipeline (Acronym Detector → Bounded Schwartz's → Character Match → Initial Capitals) followed by dictionary lookup + BiLSTM-based disambiguation model, deployed via Flask web server.
- Feature Extraction: GloVe embeddings for word representation.
- Data Preprocessing: Text tokenization with SpaCy; corpus chunking (100,000 samples per chunk) to manage the massive MAD dataset for scalable training.
- Training Method: Supervised training of separate disambiguation models per data chunk, since each acronym belongs to only one chunk.

- Evaluation Process: Benchmarked on SciAI and SciAD test sets using P/R/F1/Macro-F1.

### Student Notes

- What is the paper mainly about? A practical, publicly available web tool (MadDog) that finds acronyms in text and figures out their correct meanings across multiple domains.
- What problem does it solve? Lack of a unified, multi-domain, publicly accessible acronym identification and disambiguation tool.
- Why is this research important? It turns research findings into a usable real-world tool that can help readers understand technical documents faster.
- Main contributions: A refined rule-based AI model; a huge automatically-built glossary (DOG) and disambiguation dataset (MAD); a deployed web system.
- What to remember: Combines simple, high-precision rules with deep learning disambiguation, and covers multiple domains, unlike prior single-domain systems.

### Easy Explanation

Imagine reading a paper full of confusing short forms like "GDP" - is it "Gross Domestic Production" or something else? MadDog is a website that automatically finds these short forms in any text and tells you what they mean, even if the definition isn't given nearby. It uses smart rules to spot acronyms and their definitions when they appear together, and a trained AI model to guess the right meaning when there are multiple possibilities, based on the surrounding text. It works across many different types of text - scientific papers, Wikipedia, Reddit posts, and medical abstracts.

### One Paragraph Summary

MadDog is the first publicly available, web-based system capable of both identifying acronyms and disambiguating their meanings across multiple domains, including scientific, biomedical, general, and financial text. The system has two components: a rule-based Acronym Identification module that extends Schwartz and Hearst's classic rules with additional high-precision rules (Character Match, Initial Capitals), and an Acronym Expansion module that uses a massive glossary called DOG (426,389 acronyms, 3.78 million long forms) built from diverse corpora, combined with a BiLSTM-based deep learning model trained on an automatically labeled 46-million-record dataset (MAD) to disambiguate acronyms based on their context. Evaluated on the SciAI and SciAD benchmarks, MadDog outperforms existing rule-based, feature-based, and deep learning baselines, achieving 88.12 Macro F1 on identification and 88.49 F1 on disambiguation. The system addresses a key limitation of prior work—lack of multi-domain coverage and public availability—and demonstrates that combining refined rules with large-scale automatically labeled training data yields strong, generalizable performance. It is deployed as a live web application and open-source project, with potential future integration into e-readers.

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## Paper 4

### Paper Information

- Title: BERT-based Acronym Disambiguation with Multiple Training Strategies
- Authors: Chunguang Pan, Bingyan Song, Shengguang Wang, Zhipeng Luo
- Year: 2021
- Journal/Conference: AAAI-21 Workshop on Scientific Document Understanding (SDU); also arXiv:2103.00488 and CEUR-WS Vol-2831 proceedings
- DOI: Not Mentioned

### Summary

- Research Problem: Few works have addressed acronym disambiguation (AD) specifically in the scientific domain using pre-trained language models with multiple advanced training strategies.
- Objective: To build a high-performing binary classification model for AD that ranked 1st in the SDU@AAAI-21 Acronym Disambiguation shared task.
- Motivation: BERT and similar pre-trained language models are effective for contextual representation but need additional strategies to handle the imbalance and small size of scientific AD datasets.
- Method: Formulate AD as a binary classification task (sentence + acronym + candidate expansion → match or not). Use BERT encoder with candidate expansion and text as two input segments; concatenate the acronym's start/end mean representation with [CLS] token, pass through classifier layers. Combine multiple training strategies: dynamic negative sample selection, task adaptive pretraining (TAPT), adversarial training, and pseudo-labeling.
- Dataset: SciAD (62,441 samples from 6,786 arXiv papers).
- Algorithms: BERT, RoBERTa, cs-RoBERTa, SciBERT (compared); binary classification model with dropout, dense, ReLU, sigmoid layers.
- Evaluation Metrics: Precision, Recall, F1 (macro F1).
- Results: Final model achieved Precision 0.9695, Recall 0.9132, F1 0.9405 on test data - ranked 1st in the shared task, outperforming the baseline (GAD, F1 0.8190) by 12.15%.
- Key Findings: SciBERT as base model outperforms BERT and RoBERTa; each added strategy (dynamic sampling, TAPT, adversarial training, pseudo-labeling) incrementally improves F1; dynamic negative sampling gave the largest boost (+4%).
- Advantages: State-of-the-art results; systematic ablation of multiple training strategies; near-human performance (human = 0.9610 F1).
- Limitations: Struggles when candidate expansions are very similar in meaning (e.g., singular/plural forms) or when context provides insufficient information.
- Future Work: Not explicitly detailed beyond current comparisons.
- Applications: Text classification, question answering, scientific document understanding.

## Keywords

- Author Keywords: Not Mentioned
- Technical Keywords: Acronym Disambiguation, BERT, Dynamic Negative Sampling, Task Adaptive Pretraining, Adversarial Training, Pseudo-Labeling, SciBERT

## Technical Concepts

- Machine Learning / Deep Learning Models: BERT-based binary classifier (scibert-scivocab-uncased base).
- Architecture or Framework: BERT encoder → mean of acronym start/end token representations concatenated with [CLS] → dropout → dense → ReLU → dropout → dense → sigmoid.
- Feature Extraction: Contextual embeddings from BERT; acronym position marked with <start>/<end> tokens.
- Data Preprocessing: Candidate expansion as first input segment, sentence as second segment, separated by [CLS]/[SEP].
- Training Method: Dynamic negative sample selection per batch; task-adaptive pretraining (masked LM on task-specific data); FGM-style adversarial training; pseudo-labeling with high-confidence test predictions (>0.95) added to training set.
- Evaluation Process: Cross-validation on SciAD test set; macro-averaged P/R/F1.

## Student Notes

- What is the paper mainly about? Building a strong BERT-based binary classifier to determine whether a candidate long form correctly matches an acronym's meaning in context.
- What problem does it solve? Improving acronym disambiguation accuracy through combined training strategies rather than a single technique.

- Why is this research important? It shows how stacking several relatively simple training tricks can push a BERT model close to human-level performance on a hard NLP task.
- Main contributions: A binary classification formulation of AD; systematic combination of TAPT, adversarial training, dynamic sampling, and pseudo-labeling; 1st place shared task result.
- What to remember: Dynamic negative sampling had the single biggest performance boost among all four strategies tested.

### Easy Explanation

This paper treats the acronym-meaning matching problem as a simple yes/no question: "Does this candidate meaning match the acronym in this sentence?" They used BERT (a powerful text-understanding AI) and added several training tricks: (1) balancing how many wrong answers are shown during training, (2) pretraining BERT further on similar text before the main task, (3) adding tiny noise during training to make the model tougher, and (4) using the model's own confident guesses on test data as extra training examples. Combining all these tricks made their system the best performer in a major NLP competition.

### One Paragraph Summary

This paper proposes a BERT-based binary classification approach for scientific acronym disambiguation (AD), where the model is given a sentence, an ambiguous acronym, and a candidate expansion, and must predict whether they match. The architecture concatenates the BERT [CLS] representation with the mean of the acronym's start and end token embeddings, then passes this through dropout and feed-forward layers ending in a sigmoid output. The authors combine four training strategies—dynamic negative sample selection, task-adaptive pretraining (TAPT), adversarial training, and pseudo-labeling—built on top of a SciBERT base model, which outperformed BERT-base, BERT-large, RoBERTa-base, and cs-RoBERTa in initial experiments. Evaluated on the SciAD dataset, the final combined model achieved a test F1 of 0.9405, ranking first in the SDU@AAAI-21 Acronym Disambiguation shared task and outperforming the previous best baseline (GAD) by over 12 percentage points, approaching human-level performance (0.9610 F1). Error analysis revealed remaining challenges with near-synonymous candidate expansions and sentences with insufficient contextual information. The work highlights that systematically layering multiple, complementary training strategies onto a strong pre-trained language model can yield state-of-the-art results on a challenging, ambiguous NLP disambiguation task.

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## Paper 5

### Paper Information

- Title: Leveraging Domain Agnostic and Specific Knowledge for Acronym Disambiguation
- Authors: Qiwei Zhong, Guanxiong Zeng, Danqing Zhu, Yang Zhang, Wangli Lin, Ben Chen, Jiayu Tang
- Year: 2021
- Journal/Conference: AAAI-21 Workshop on Scientific Document Understanding (SDU); also arXiv:2107.00316 and CEUR-WS Vol-2831 proceedings
- DOI: Not Mentioned

### Summary

- Research Problem: General-domain pre-trained language models achieve insufficient performance when applied directly to scientific-domain acronym disambiguation, while scientific-domain models alone may lack fine-grained general knowledge; obtaining large annotated scientific data is expensive.

- Objective: To combine both domain-agnostic (general) and domain-specific (scientific) pre-trained knowledge to improve acronym disambiguation performance.
- Motivation: Neither purely general nor purely domain-specific pre-trained models alone provide the best representation for acronym disambiguation in scientific text; combining both could capture complementary information.
- Method: Propose hdBERT (Hierarchical Dual-path BERT): RoBERTa (via Byte-Pair Encoding) captures general fine-grained semantic information, and SciBERT (via WordPiece) captures high-level scientific domain information. Both are used to encode input (sentence + candidate long form), and their representations are concatenated and fed into a multi-layer perceptron (MLP) with sigmoid output for binary classification.
- Dataset: SciAD (62,441 sentences, 732 ambiguous acronyms), converted into a binary classification dataset SciAD\_BI (352,366 train / 28,286 dev / 28,364 test after upsampling).
- Algorithms: RoBERTa, SciBERT, BERT (compared individually); hdBERT (proposed ensemble via MLP).
- Evaluation Metrics: Macro Precision, Macro Recall, Macro F1.
- Results: hdBERT achieved Macro F1 of 93.73%, outperforming BERT (90.90%), RoBERTa (92.00%), and SciBERT (92.95%); GAD baseline only achieved 81.90%. Human performance is 96.10%.
- Key Findings: Combining general (RoBERTa) and domain-specific (SciBERT) representations yields better performance than either alone; SciBERT alone outperforms domain-agnostic models, confirming importance of scientific pretraining; the loss curve of hdBERT converges faster and lower than single models.
- Advantages: Effectively fuses complementary knowledge sources; state-of-the-art results among compared methods; simple MLP-based integration.
- Limitations: Still below human performance, especially on macro recall; struggles with certain hot/trendy phrase confusions (e.g., predicting "Federated Learning" instead of correct meaning) and noisy conflicting annotations in SciAD.
- Future Work: Explore self-training, adversarial learning, and contrastive learning to further improve performance.
- Applications: Scientific document understanding, automated glossary/definition systems.

## Keywords

- Author Keywords: Not Mentioned
- Technical Keywords: Acronym Disambiguation, hdBERT, RoBERTa, SciBERT, Dual-path BERT, Domain Agnostic Knowledge, Domain Specific Knowledge, Word Sense Disambiguation

## Technical Concepts

- Machine Learning / Deep Learning Models: RoBERTa (BPE-based), SciBERT (WordPiece-based), combined via Multi-Layer Perceptron (MLP).
- Architecture or Framework: Hierarchical Dual-path BERT (hdBERT) — parallel RoBERTa and SciBERT branches whose outputs are concatenated and passed through an MLP with sigmoid activation.
- Feature Extraction: Byte-Pair Encoding (RoBERTa) and WordPiece Encoding (SciBERT) for tokenization/embedding.
- Data Preprocessing: Converted multi-class AD problem into binary classification (sentence + acronym + candidate → match/no match); upsampling of positive samples to balance dataset.
- Training Method: Cross-entropy loss with L2 regularization; Adam-based fine-tuning of both branches jointly.
- Evaluation Process: Macro Precision/Recall/F1 on SciAD test split; comparison against multiple baselines.

## Student Notes

- What is the paper mainly about? Combining two types of BERT models (general-domain RoBERTa and scientific-domain SciBERT) to better disambiguate acronyms in scientific text.
- What problem does it solve? The trade-off between general language understanding and domain-specific scientific knowledge for acronym disambiguation.
- Why is this research important? It shows that combining complementary pre-trained models can outperform using either type alone.
- Main contributions: First attempt to jointly leverage domain-agnostic and domain-specific knowledge for AD; introduction of hdBERT architecture; strong empirical results.
- What to remember: hdBERT's core idea is "two experts are better than one" — merging fine-grained general knowledge with scientific domain expertise.

### Easy Explanation

Some AI models (like RoBERTa) are trained on huge amounts of general text and are good at understanding language in general, while others (like SciBERT) are trained only on scientific papers and understand science jargon better. This paper combines both types of models together - like having two experts vote on the answer - to better guess what a confusing acronym in a science paper means. The combined model, called hdBERT, performed better than using either expert alone, achieving very high accuracy on a large disambiguation dataset.

### One Paragraph Summary

This paper proposes hdBERT (Hierarchical Dual-path BERT), a model for scientific acronym disambiguation that simultaneously leverages both domain-agnostic and domain-specific knowledge. The architecture uses two parallel branches: RoBERTa, pretrained on general-domain corpora via Byte-Pair Encoding, captures fine-grained general semantic information, while SciBERT, pretrained on scientific text via WordPiece tokenization, captures high-level domain-specific knowledge. Both branches process the same input (a sentence containing an ambiguous acronym plus a candidate long form), and their outputs are concatenated and passed through a multi-layer perceptron for binary classification of whether the candidate is correct. Evaluated on the SciAD dataset (converted to a binary classification format, SciAD\_BI), hdBERT achieved a macro F1 of 93.73%, outperforming standalone BERT (90.90%), RoBERTa (92.00%), and SciBERT (92.95%), as well as earlier non-BERT baselines like GAD (81.90%). The model's loss curve also converged faster and lower than single-model baselines, confirming the benefit of fusing complementary representations. Despite these gains, performance remains below human-level accuracy (96.10% F1), with error analysis showing persistent challenges from semantically similar candidate meanings and noisy conflicting labels in the dataset. The authors suggest future work incorporating self-training, adversarial learning, and contrastive learning to close this gap further.

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## Paper 6

### Paper Information

- Title: Primer AI's Systems for Acronym Identification and Disambiguation
- Authors: Nicholas Egan, John Bohannon
- Year: 2021
- Journal/Conference: AAAI-21 Workshop on Scientific Document Understanding (SDU); also arXiv:2012.08013 and CEUR-WS Vol-2831 proceedings
- DOI: Not Mentioned

### Summary

- Research Problem: Existing acronym identification (AI) and disambiguation (AD) models can be improved by leveraging additional distantly-supervised data, and the SciAD dataset itself has quality issues (duplicate examples with conflicting labels) that bias evaluation.
- Objective: To propose improved AI and AD systems, release new auxiliary datasets, and identify/fix a duplication problem in SciAD.
- Motivation: Better performance and data quality are needed to advance the SDU@AAAI-21 shared tasks; addressing SciAD's duplicate/conflicting labels problem is essential for fair evaluation.
- Method: For AI, use XLNet (transformer-based language model) to embed tokens, then project embeddings onto BIO tag logits via a linear layer; ensemble multiple XLNet models with different hyperparameters; optionally pretrain on a new distantly-supervised dataset (AuxAI) before finetuning on SciAI. For AD, reframe the task as information retrieval: train Twin/Siamese Networks (transformer encoders) to produce sentence embeddings such that sentences sharing the same acronym meaning have higher cosine similarity; at inference, find the most similar training sentence and use its label; ensemble multiple pretrained sentence-transformer models (SIF, XLM paraphrase, DistilRoBERTa variants, DistilBERT Quora, RoBERTa SciAD, XLM+AuxAD).
- Dataset: SciAI, SciAD (from Veyseh et al.); new datasets: AuxAI (313,914 distantly-supervised sentences), AuxAD (112,788 examples), and SciAD-dedupe (deduplicated version of SciAD).
- Algorithms: XLNet (AI); Twin/Siamese Networks with SIF, XLM, DistilRoBERTa, DistilBERT, RoBERTa encoders (AD).
- Evaluation Metrics: Precision, Recall, F1 (macro-averaged).
- Results: AI: XLNet ensemble achieved F1 of 93.63 (dev) / 92.60 (test), outperforming LSTM-CRF baseline (86.55). AD: Ensemble achieved F1 of 91.22 (SciAD dev) / 93.15 (SciAD-dedupe dev) / 91.58 (test), outperforming the GAD baseline (81.90).
- Key Findings: Pretraining on distantly-supervised AuxAI before finetuning on SciAI improves performance; SciAD contains significant duplicate (sentence, acronym) pairs (12,672 of 62,441) with some conflicting labels, upper-bounding achievable accuracy at ~93.1% on duplicated subset; deduplication (SciAD-dedupe) provides a fairer benchmark.
- Advantages: Novel information-retrieval framing for AD; identification and correction of a significant dataset quality issue; strong ensemble performance.
- Limitations: AuxAI alone performs poorly (F1 66.96) due to less diverse acronym coverage; approach depends on having similar training examples available (retrieval fails without good matches).
- Future Work: Not explicitly detailed beyond noting potential for larger training corpora to address retrieval failures.
- Applications: Scientific document understanding, information retrieval-based disambiguation systems.

## Keywords

- Author Keywords: Not Mentioned
- Technical Keywords: Acronym Identification, Acronym Disambiguation, XLNet, Twin/Siamese Networks, Sentence Embeddings, AuxAI, AuxAD, SciAD-dedupe, Information Retrieval

## Technical Concepts

- Machine Learning / Deep Learning Models: XLNet (transformer-based LM) for token tagging; Twin/Siamese sentence-transformer networks for embedding-based retrieval.
- Architecture or Framework: AI: XLNet encoder → linear projection to BIO tag logits, ensembled across hyperparameter configurations. AD: Siamese encoder network trained via cosine similarity and MSE loss to embed sentences such that same-meaning sentences cluster together; nearest-neighbor retrieval used at inference.

- Feature Extraction: SentencePiece byte-pair encoding tokenization; sentence embeddings from various pretrained sentence-transformer models.
- Data Preprocessing: Distant supervision using Abbreviations.com scraped abbreviation-expansion pairs matched against arXiv abstracts (for AuxAI/AuxAD); deduplication of SciAD by removing repeated (sentence, acronym) pairs and resolving conflicting labels via majority vote.
- Training Method: Cross-entropy loss + AdamW optimizer (AI); Mean Squared Error loss on cosine similarity for Siamese Networks (AD).
- Evaluation Process: Macro-averaged F1 on SciAI dev/test; macro F1 on SciAD and SciAD-dedupe dev/test sets; error analysis via manual review of 50 random mistakes.

## Student Notes

- What is the paper mainly about? Two improved systems—one for identifying acronyms (using XLNet) and one for disambiguating them (using sentence similarity/retrieval)—plus new auxiliary datasets and a fix for a data quality problem in SciAD.
- What problem does it solve? Improves accuracy on both AI and AD tasks and addresses hidden data leakage/duplicate label issues in the SciAD benchmark.
- Why is this research important? Shows how distant supervision and creative task reframing (disambiguation as retrieval) can improve results, and highlights the importance of clean benchmark data.
- Main contributions: XLNet-based AI ensemble; Twin Network-based AD retrieval approach; three new public datasets (AuxAI, AuxAD, SciAD-dedupe).
- What to remember: The SciAD dataset had thousands of duplicate examples with conflicting labels, meaning some reported "high accuracy" was partly due to data leakage, not true model skill.

## Easy Explanation

Primer AI built two systems: one to spot acronyms in text using a strong language model called XLNet, and another to guess what an acronym means by finding a similar sentence from past examples and copying its answer (like "find the closest matching example and use its label," similar to searching a big lookup table intelligently). They also noticed something important: the SciAD dataset used by many researchers had lots of exact duplicate sentences appearing in both the training and test sets, sometimes even with different "correct" answers. This meant models could look like they were doing great just by memorizing duplicates, not by actually understanding the meaning. They fixed this by making a cleaned-up version of the dataset, SciAD-dedupe, for fairer testing.

## One Paragraph Summary

This paper presents Primer AI's approach to the SDU@AAAI-21 shared tasks on acronym identification (AI) and disambiguation (AD). For AI, the authors use an XLNet-based transformer model with a linear projection to BIO tag logits, boosted by pretraining on a new distantly-supervised dataset (AuxAI, 313,914 sentences) built by scraping Abbreviations.com and matching terms in arXiv abstracts; an ensemble of 15 XLNet models achieved an F1 of 92.60 on the test set, well above the LSTM-CRF baseline. For AD, they reframe the task as information retrieval: Twin (Siamese) Networks are trained to embed sentences so that those sharing the same acronym meaning are close in cosine similarity space, and predictions are made by copying the label of the nearest training example; an ensemble of several pretrained and fine-tuned sentence-transformer models achieved an F1 of 91.58 on the test set. Critically, the authors discovered that the widely-used SciAD dataset contains substantial duplication—12,672 of 62,441 examples share identical (sentence, acronym) pairs with training data, and some duplicates have conflicting labels, capping achievable accuracy and biasing evaluation. To address this, they released SciAD-dedupe, a deduplicated version of the dataset, alongside AuxAI and AuxAD as new public resources. This work both advances model performance through creative reframing (retrieval-based AD) and improves benchmark data quality for the scientific document understanding community.

